

Examiner reports

Q1.

Many students were able to score at least 2 marks by rearranging the given equation and finding at least three correct solutions. Fewer students considered both the negative and positive square roots of 3 to obtain the full set of 8 correct answers. The majority of students made a significant error rearranging the given equation, $\tan^2 2\theta - 3 = 0$ becoming $\tan^2 \theta = \frac{3}{2}$ was the most common error. Attempts made by expressing $\tan^2 2\theta$ in terms of $\sin^2 2\theta$ and $\cos^2 2\theta$ were rarely successful.

Q4.

This final question resulted in a wide spread of marks. In part (a) a significant number of students did not see that the answers could be written down directly. The most common incorrect method led to incorrect equations similar to $x + 30 = 89.27$, the right-hand side coming from $\tan^{-1}(79)$. Most students correctly recognised the geometrical transformation in part (b) to be a translation but the direction of the translation was quite frequently incorrect. In part (c)(i) the right-hand side of the given equation was usually expanded correctly with the correct notation seen and many recalled and used the correct relevant trigonometric identity to eliminate $\sin^2 \theta$. However, solutions towards the printed answer were spoilt in a significant minority of cases by students incorrectly writing the correct expression $5 + 1 - \cos^2 \theta$ as $5 - 5\cos^2 \theta$. Of those who obtained the correct simplified quadratic equation in $\cos \theta$, most were able to solve the equation correctly but not all of these students gave an indication that they had considered and rejected the value -2 for $\cos \theta$. The final part of this last question proved to be a challenge even for the more able students. Many solutions just consisted of solving $\cos \theta = \frac{3}{4}$ to obtain two solutions rather than stating the required equation as $\cos 2x = \frac{3}{4}$ and solving it to find the four solutions within the given range to the required degree of accuracy.

Q5.

In Part (a), most students gained the method mark by using the identity $\sec^2 x = 1 + \tan^2 x$ and many went on to earn the accuracy marks. A significant number of students lost the final accuracy mark through poor notation such as writing $\tan x^2$ instead of $\tan^2 x$. It was surprising to see some students working at this level to write $\frac{\sec^2 x}{\sec^2 x - 1}$ as $\frac{\sec^2 x}{\sec^2 x} - \frac{\sec^2 x}{1}$.

Part (b) proved difficult for most students. Many students assumed that they could factorise the quadratic equation and they were satisfied with factors that were clearly incorrect, such as $(\operatorname{cosec} x - 3)(\operatorname{cosec} x + 1)$. Some students misquoted the quadratic formula; some others muddled factors and roots. Most students showed a correct conversion from $\operatorname{cosec} x$ to $\sin x$ after obtaining their roots but those who attempted to convert their equation at the start rarely proceeded to a correct quadratic equation in $\sin x$. Some students discarded the negative root of their quadratic equation.

In part (c), the students who answered part (b) correctly usually gained both marks for this

part, although some lost the accuracy mark for not writing their answers to the nearest degree. Where both marks were not earned, the majority of the students gained the method mark for equating $2\theta - 60^\circ$ to one of their solutions in part (b).

Q6.

The majority of students did well on part (a) of this question and well over half of the cohort scored 6 marks or more. Most students were less successful on part (b), although many started by writing down several relevant looking identities.

Part (a)(i). The vast majority of students answered correctly, with relatively few finding $\alpha = 56.3^\circ$ rather than the expected $\alpha = 33.7^\circ$.

Part(a)(ii). Although most students scored the 3 marks some failed to state the minimum value where an explicit statement was required. Similarly, some students just wrote down the value of x at which the minimum occurred without justifying it. Less common errors

were to use $\cos^{-1}\left(-\frac{1}{R}\right)$ or to assume the minimum was at $x = 90^\circ$.

Part (b)(i). Many students wrote the given expression in terms of $\sin x$ and $\cos x$, but then went on to treat the identity as if it were an equation, often equating to zero. Those who used a common denominator or took out a factor of $\frac{1}{\sin x}$ usually completed successfully.

Part (b)(ii). Although most students recognised that they had to use the result from part (i), many failed to consider both $\cot x = 0$ and $\cos 2x = 0$ and so gained no credit. Of those who did consider both these equations many students thought $\cot x = 0$ had no solutions, or they changed it to $\tan x = 0$. Most students did solve $\cos 2x = 0$ correctly, although not always giving both solutions within the given range.

Q7.

Sketching a convincing tangent graph, even with graphical calculators available, seemed a considerable challenge for many. Three separate branches were needed with the correct intercepts on the x -axis and demonstration of the tendency near 90° and 270° . It was disappointing to sometimes see a sine or a cosine graph drawn instead. Many students correctly solved the trigonometric equation in part (a)(ii) although a minority of students incorrectly gave more than two solutions inside the given interval. The majority of students were able to recall and use the required two trigonometric identities in part (b)(i) but not all were able to reach the printed answer without there being a sign error in their working. In the final part of this last question on the paper it was important that students recognised that they were required to use 'Hence'. and also that the equation to be solved was in terms of $3x$. Those students who just solved the equation in part (b)(i) without ever dividing its solutions by 3 or writing ' $\theta = 3x$ '. were heavily penalised. Those students who did link θ to $3x$ generally recognised and solved the quadratic in $\cos 3x$ but a significant minority then failed to find a third solution for x in the given interval.

Q8.

This was a very well answered question with a high proportion of students gaining full marks. A small number of students failed to round results to three significant figures or gave their answers in degrees. Some lost marks through sign errors when collecting the terms of the quadratic together prior to factorisation. The most fundamental error seen

was where students tried to write everything in terms of $\tan x$ and occasionally the equation $8(1 + \tan x) - 2(1 + \tan^2 x) = \tan^2 x - 2$ was seen. Those who changed to sine and cosine at the start had to convert to an equation all in cosine to earn marks and some did not go far enough.

Q9.

This question was generally done well with about 35% of the entry gaining 10 or 11 marks. Often the mark lost was in part (a) for failure to show any processing or to give a conclusion.

- (a) Virtually all students knew they were to substitute $x = -\frac{3}{2}$, and evaluate. However, those who just substituted into $f(x)$ and wrote $= 0$ without showing any stages in the arithmetic did not score the accuracy mark, nor did those who did not interpret the $= 0$ result.
- (b) Most students had this correct and virtually all scored at least the method mark. Some students did the factorisation by inspection whilst long division was also a popular method. Those students who chose to use undetermined coefficients often did much unnecessary work, as at least the values of a and c should be apparent from inspection.
- (c) (i) This part of the question was also generally done well with most students able to quote the cosine double angle formula or derive it correctly, and convert it to sines.
- A rare error was a sign error or the dropping of the 2. Some students did not handle multiplying out the expression convincingly, and as the required result was given, they were penalised. However, most did convert their trigonometric expression into the required equation in x .
- (ii) Some students did not attempt to solve this equation, but of those who did they virtually all knew they had to solve the cubic equation in x . Many did this successfully, often following a failed attempt at factorisation. Some made sign or other errors in their use of the quadratic formula. Some students used the quadratic formula on the cubic equation itself. Students who had not used the quadratic formula correctly with the given equation, could gain no further marks. However, many did complete successfully and it was pleasing to see the inappropriate roots of the cubic equation carefully rejected. Some students did not correctly convert the -16.3° they obtained from their calculator into angles in the required range.

Q10.

In part (a), it was clear that many candidates could recall the relevant identity, but rearrangement of $2 \sin \theta = 7 \cos \theta$ into a correct form for applying the identity caused more difficulty than usual.

In part (b)(i), most candidates scored the method mark, but insufficient care in reaching the acceptable form of the printed answer resulted in the loss of the accuracy mark; for example, answers which involved a mixture of x s and θ s were seen, as well as those which omitted the ' $= 0$ '. In part (b)(ii), almost all candidates realised the need for using

part (b)(i) and reached the correct quadratic equation. However, poor factorisation or failing to realise that the roots of the equation after applying the quadratic formula represented the values of $\cos x$, rather than x , led to a significant loss of marks. There were however a substantial number of fully correct solutions which displayed a much better understanding of this type of question than has been the case in similar questions in a number of previous papers.

Q11.

A few candidates made no attempt at this question at all.

Part (a) was very well answered by the majority of candidates, with many obtaining full marks. Most candidates used the correct substitution for $\tan^2 \theta$ and went on to produce the correct quadratic function. Factorisation was usually handled correctly with the common error of

$\sec \theta = -2$ or 5 occasionally seen. The major error noticed was with the answers produced: 60° , 300° , 101.5° were usually correct but the fourth value was often given as 281.5° (from $101.5^\circ + 180^\circ$), rather than the correct solution of $180^\circ + 78.5^\circ = 258.5^\circ$.

In part (b), although most candidates recognised the connection with part (a), not all realised what 'hence' implied. Many gave fully correct answers, but it was disappointing, at this level, how many candidates got the algebra wrong by dividing by 4 first and then adding 10. However, a significant number of candidates chose to start again and most were unable to handle the new expression and gained no further marks.

Q12.

In part (a), virtually all candidates knew that they were to substitute $x = \frac{3}{4}$ into the cubic expression, but many either failed to show any detail of a calculation leading to zero and / or failed to give a conclusion as to the significance of such a result. Candidates need to recognise that when a printed answer is given, as with $(4x - 3)$ being a factor here, they need to give a full justification of why this is the case.

In part (b), most candidates showed that they were familiar with the double angle formulae, but more so the double sine formula, with there often being an error in that of the double cosine. However, many substituted correctly, and manipulated their expression in θ correctly using $\sin^2 \theta = 1 - \cos^2 \theta$, before converting to x . Some substituted x for $\cos \theta$ as soon as they could and sometimes got into a confused expression which they were unable to deal with convincingly. Some candidates tried to approach the problem by working on the sine and cosine parts separately; sometimes they were successful but again some became confused when they tried to put it back together in terms of x . Some candidates failed to secure the final mark by not equating to zero.

Most candidates attempted part (c) and knew that they were to factorise the cubic expression. Some used long division by the given factor of $(4x - 3)$ but then did nothing with their resulting quadratic expression, whereas others showed the correct interpretation and wrote down the factorised form of the cubic expression. Some candidates factorised the cubic expression by inspection, mostly correctly, but with the occasional error in the x -term of the quadratic factor. Some candidates who had a correct factorisation failed to score the last two marks as they simply said that the quadratic did not factorise or that it had no roots, without demonstrating why. To score the marks, candidates were required to calculate their discriminant and, if correct, note that it was negative and thus that there

were no more (real) solutions to the cubic equation. Most of those who did this concluded the question with a convincing argument as to why there was only one solution to the cubic equation and thus to the trigonometric equation and stated the real solution.

Q13.

This question on trigonometric equations and identities was by far the least-well-answered question on the paper. The equation was given in a factorised form (in the hope that students would be led to write down $\tan \theta + 1 = 0$, $\sin^2 \theta = 3\cos^2 \theta$) but this was overlooked by many students who chose to multiply out, and create a more complicated problem from which recovery was unusual. Even those who separated the two factors,

and divided the second by $\cos^2 \theta$, often 'cancelled' $\frac{\sin^2 \theta}{\cos^2 \theta}$ erroneously to $\tan \theta$. Of those who overcame the initial hurdles, a substantial number ignored the possibility of a negative square root and so missed the value $-\sqrt{3}$ for $\tan \theta$.

Q14.

This question was poorly answered.

In part (a), although many students achieved full marks, some found the combining of two fractions beyond them. For many students, an initial step was replacing $1 + \cos \theta$ with $\sin \theta$ or attempting to multiply through by $1 + \cos \theta$, thus obtaining expressions such as

$$1 + \frac{1 + \cos \theta}{1 - \cos \theta} = 32$$

Despite the majority of students finding part (b) of this question difficult, full marks were occasionally seen. Students who were unsuccessful in part (a) were able to attempt this part using the result given. Most students earned the first method mark, but for many this was all they achieved since they only worked with $\sin \theta = +\frac{1}{4}$ and then only worked to 2dp which resulted in just 2 solutions. For the final two B marks, the solution $x = 0.17$ was often missing, and 2.00 was often written as 1.99.

Q15.

In part (a), whilst most students found the value of R correctly and left it as $\sqrt{10}$, the same was not true of the angle α . Some students seemingly failed to notice that α was given as an acute angle and found $\tan \alpha$ to be -3 ; some confused sine and cosine and had $\tan \alpha = \pm \frac{1}{3}$. Some students progressed to $\tan \alpha = 3$ from $\sin \alpha = 3$, $\cos \alpha = 1$ which was penalised for both marks.

For part (b), most students rearranged the equation correctly using their result from part (a), although many then dropped the negative sign when finding the inverse sine. Many students seemingly could not cope with the result of -39.23° that their calculator gave them, when all that was required was to add 71.6° (ie α) to get the first solution. Some students found the third quadrant solution only, whereas some found more than two solutions. There were many students who ignored the request to round to the nearest degree in their final answer; this was condoned if their answers were correct to the greater number of decimal places they gave. Many students did give a fully correct solution, often by using a sketch graph to help them decide where the solutions lay.

Q16.

Part (a) was generally answered very well although some candidates only gave one value within the range despite the clue in the question '....your answers to...' or gave values in the wrong two quadrants.

Part (b)(i), as expected, proved to be the most demanding part question on the paper. The two most common successful initial steps seen were either replacing the 6 on the right-hand side of the equation by $6(\sin^2 \theta + \cos^2 \theta)$ or splitting the $7\sin^2 \theta$ into $\sin^2 \theta + 6\sin^2 \theta$ and then rearranging to get $\sin^2 \theta + \sin \theta \cos \theta - 6(1 - \sin^2 \theta) = 0$. However, for most candidates the maximum mark awarded was 1 for correctly using a relevant trigonometrical identity. Many candidates tried more than one approach, all of which were usually unsuccessful whilst many, having (wrongly) reached the stage $7\tan^2 \theta + \tan \theta = 6$ appeared happy to lose the 7, rearrange and claim the printed result.

It was pleasing to see most of those candidates who attempted part (b)(ii) using the quadratic in $\tan \theta$ printed in part (b)(i) to start their solution. There were many correct solutions presented, although poor factorisation was seen in some attempts. It was surprising to see some candidates using their own version of the quadratic, $7\tan^2 \theta + \tan \theta - 6 = 0$, in part(b)(ii) and this gained no credit.

Q17.

In part (a) most candidates were able to obtain the first angle of 1.77 radians and hence obtain 2 marks. Many candidates also obtained the correct second solution, although 4.91 was a common error.

Part (b) was the worst answered question part on the paper, with very few candidates having any idea of how to go about combining the two fractions. Very few completely correct solutions

were seen. Those who managed to reduce the LHS to $\frac{-2\operatorname{cosec}^2 x}{-\cot^2 x}$ could often not complete the question. It was sad to see some very poor algebraic techniques in evidence here. There were many cases where candidates did not attempt the question.

In part (c) far too many candidates failed to recognise both the positive and the negative roots for $\sec x$ and in consequence they did not score any marks, since they only found the solutions for $\cos x = 0.2$.

Q18.

There was a wide range of responses to this question and relatively few candidates gained full marks.

In part (a)(i) most candidates wrote down a correct form for $\tan 2x$ with only a few writing $\tan^2 x$ instead of $2\tan x$. Most multiplied by the denominator to obtain a cubic equation in $\tan x$. However, many divided throughout by $\tan x$ to get $\tan^2 x = 3$ and failed to demonstrate that $\tan x = 0$ is also a solution. Some used verification which did not warrant any marks.

In (a)(ii) very few candidates gained this mark. The solution of 120° was almost always

omitted. Most wrote 0° and 180° too. This in itself was not penalised this time as these values were outside the range of x requested.

Part (b)(i) was answered very well, as candidates showed familiarity with double angle formulae and handled the algebra involved with confidence.

In part (b)(ii) the quadratic formula was used correctly by most candidates and most demonstrated where the $\sqrt{3}$ came from. The majority of candidates, though, just seemed to ignore the negative root and gave no explanation as to why. If they found only one root then they lost 2 marks. Some gave unsatisfactory explanations for the exclusion of the negative root, such as $\sin x$ cannot be negative or that it was > 1 . Those who used completion of the square to solve the equation almost invariably just took the positive square root.

Q19.

Part (a) was answered well with most candidates using the trapezium rule with the correct number of strips and values substituted correctly. The most common errors were using degrees instead of radians and incorrect use of brackets.

Most candidates recognised that the required transformation for part (b) was a stretch, but were less sure of the direction and/or scale factor.

In part (c) the most common approach was to attempt to write the given equation in terms of

$\tan x$ with varying degrees of success. Although some very good solutions were seen, a common wrong method is illustrated by ' $2\tan x = 1$, $2x = 0.785$, 3.93 , $x = 0.393$, 1.97 ,'

Those who correctly reached either $\sin^2 x = \frac{1}{5}$ or $\cos^2 x = \frac{4}{5}$ rarely scored more than one of the four marks because they forgot to consider both the positive and negative roots of these equations.

Q20.

In part (a)(i), although the majority of candidates scored both marks here some rounded to the nearest degree, some truncated, some rounded 14.477 to 14.8, some only considered +4, and

some erroneously added -14.5 or 14.5 to 90 . A handful took cosec to be $\frac{1}{\tan}$ or $\frac{1}{\cos}$.

In (a)(ii) there were some excellent full solutions here but also some disturbing misunderstanding of notation. Many candidates found it disconcerting to deal with the functions of $(2x + 30)$, for example $2\operatorname{cosec}^2 x - 1 + (2x + 30) = 2 - 7\operatorname{cosec}(2x + 30)$ was seen. They would have been well advised to use a substitution such as Y for $(2x + 30)$. A few missed the brackets around $(1 - \operatorname{cosec}^2 Y)$ thus getting a wrong equation and no further marks; some had this correct but failed to deal with the constants correctly and again could not progress. Sadly the incorrect substitution of $\cot Y + 1$ for $\operatorname{cosec} Y$ was also seen on several occasions. Once the correct factors were obtained a few stopped there (particularly those who unfortunately substituted x for $2x + 30$). Quite a few handled the

–4 solution well but not the $\frac{1}{2}$ as they went on to assume $\sin(2x + 30)$ was $\frac{1}{2}$ and not 2.

In part (b) most candidates recognised that the required transformations were a stretch and a translation. The majority started with the stretch and then, wrongly took the translation to be $[-30, 0]$ instead of $[-15, 0]$. A few missed the correct term “stretch” and

many failed to use the right terminology of “scale factor” $\frac{1}{2}$. A few had the direction of the stretch in the y -axis or the SF as 2 and a few had their translation in the y -direction.

Q21.

This trigonometry question was the worst answered question on the paper. A common error in part (a) was to either use the wrong identity $\tan(x + 52) = \tan x + \tan 52$ or to solve $x + 52 = \tan^{-1} 22$. Although many candidates scored a mark in part (b)(i) for recalling the

identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$, a significant minority of these candidates could not recall or apply the identity $\sin^2 \theta = 1 - \cos^2 \theta$ correctly.

In part (b)(ii), many candidates applied a correct method to solve the quadratic equation but a significant number of candidates gave two values for $\cos q$ as their final answer, even though the question asked for ‘the value’. It was disappointing to see candidates attempting part (b)(iii), which started with the word ‘Hence’, without linking their solution to previous work. Although some excellent solutions to this final part were seen, the majority of candidates failed to make any real progress.

Q22.

Part (a) was reasonably well answered, with most candidates obtaining both values. Some candidates lost the accuracy mark through inaccurate evaluation of the second

angle, with 5.94 being a common incorrect answer. A few cases of $\operatorname{cosec} x = \frac{1}{\cos x}$ were seen.

Part (b) was answered very well, with most candidates who obtained full marks in part (a) also obtaining full marks in part (b). The majority of candidates earned the first 4 marks but some then lost the final mark(s) through inaccurate values. There were a few candidates who started with the wrong identity and hence scored zero.

Q23.

In part (a), most candidates were able to find R and α correctly, with only a few arithmetic

errors seen. Some candidates made the error of using $\tan \alpha = \frac{1}{3}$, or equivalent, although most candidates did use the tangent to find the value of α . Some ignored the request for three decimal places, but two decimal place accuracy was condoned.

Many candidates obtained a correct answer to part (b), although some did not give the minus sign on $\sqrt{10}$, or thought 0 or -1 was the minimum value. There was confusion among some of the candidates between the minimum value and the value of x at which it occurred, although again many correct answers were seen, but here too not always to

three decimal place accuracy.

In part (c), candidates who realised that they were to use part (a) usually made some progress. Those relatively few candidates who did not realise this usually tried to square the equation, but many did not realise this could only be successful if the sin or cos term was moved to the other side first. Many such attempts were abandoned. Those who used the expected method often lost 1 or 2 marks, through not giving all possible solutions in the range, and in particular the last accuracy mark was often lost as candidates did not make use of -0.886 in their working. Three decimal place accuracy was required for the final mark as were answers in radians: degrees were condoned during the working and in part (b) but not in part (a).

Q24.

This question, which tested trigonometric-graphs, identities and equations, was better answered than similar questions in some recent series. In part (a) most of the graphs presented had the correct shape with intercepts given, although the intercept $y = 1$ was occasionally omitted and those on the x -axis only expressed in degrees. Part (b)(i) usually had enough to justify the given result which was then generally used in part (b)(ii). The most common error in the final part was to omit the second answer for x . It was not uncommon to see weaker candidates writing 'identities' such as ' $\cos 2x \times \cos 2x = \cos^2 4x$ '.

Q25.

- (a) The majority of candidates drew a modulus graph and therefore earned the method mark. Many correct responses to the number of sections were seen although some candidates appeared to try to sketch $y = |\sec \frac{1}{2}x|$. Most candidates who earned the first two marks went on to earn the final A mark although poor sketching did lose the A mark for variable heights in some cases.
- (b) This was very well answered, with most candidates obtaining both marks. Where marks were lost it was usually through only offering the answer of 60° . A few candidates thought $\sec = 1/\tan$ or $1/\sin$ and hence obtained no marks.

Q26.

- (a)
 - (i) The first part of this question was reasonably answered with many candidates obtaining both marks. Where candidates obtained 1 mark it was because many ended up with the answer $1/(5x - 2)$.
 - (ii) Again this part was very well answered with candidates obtaining both marks. The majority of candidates arrived at $K \cos 2x$ but $K = -2$ was a common error.
- (b)
 - (i) This part was not answered very well. Many candidates lost a mark through using -0.693 instead of $\ln \left(\frac{1}{2} \right)$ and $f(x) \geq 0$ was a common response.
 - (ii) Most candidates were able to do this part with the correct answers often seen. The main error was the omission of brackets around $5x - 2$ obtaining $\sin 2(\ln 5x - 2)$. The expression for $fg(x)$ was also often seen.
 - (iii) For those candidates with a correct starting expression many went on to get

full marks. Those candidates who used $\sin \ln (5x - 2)^2$ often lost an accuracy mark for not rejecting one of their answers. Most candidates obtained the first method mark for making the correct initial step for their expression.

- (iv) This was usually well done but there were common errors of dividing by $\sin 2$ obtaining $y/\sin 2 = x$ or even $y/\sin = 2x$ and $y/2 = \sin x$

Q27.

- (a) Very well answered by the majority of candidates. Most candidates used the correct identity and were successful in answering this part of the question.
- (b) Most candidates were successful in answering this part of the question. When answered correctly the factorisation of the resulting quadratic or use of the quadratic formula was usually well done but there were some sign errors. Many totally correct solutions were seen.

Q28.

- (a) (i) Most candidates either knew the appropriate identity for $\cos 2x$ to use, or worked it out in terms of $\sin x$ from $\cos^2 x - \sin^2 x$. The common error was to attempt to use an "identity" $\cos 2x = \sin^2 x - 1$ or similar, and candidates did gain credit for attempting to get an expression in $\sin x$. Most candidates, who had used a correct identity, went on to convincingly obtain the given equation. Others, who had an incorrect identity, chose to ignore inconsistencies and just wrote down the given result. Those few candidates who had "invented" an identity that led to the result by working backwards from the answer given, gained no credit.
- (ii) Most candidates used factorisation correctly to get the required results. Those who attempted to use the quadratic formula were more prone to making an error. Some candidates attempted to solve their equation from part (a)(i) rather than the given equation, and gained no credit. Many candidates wasted some time here by proceeding to find the angles, which had not been asked for.
- (b) (i) The technique for finding R and α was well known and most candidates did this correctly. The common error was to find α as 56° , and less so a numerical error in finding R . Some candidates appeared not to notice they were given $R > 0$, and gave $\pm R$ in their result. Another error sometimes seen was for candidates to confuse $2x$ and α , and thus get half the expected value for α .
- (ii) Although most candidates knew they were to use their result from part (b)(i) to solve this equation, many made an error in manipulating the given equation to a soluble form. Either 0 or 1 appeared on the right hand side rather than -1 and such candidates could gain no further credit. Whilst many candidates proceeded to solve the equation for one correct angle, relatively few got both correct and some got neither correct. Those who drew a sketch of the cosine curve, usually found it helpful in realising there were two solutions. Those few candidates who did not attempt to use the result from part (b)(i) to solve the equation gained no credit. Attempts just by using a calculator were very rare, and there had to be evidence of using part (b)(i) to gain any credit.

Q29.

Many candidates found the inverse sine of 0.8 correctly but a significant minority did not give the second solution. Parts (b)(i) and (b)(ii) were usually well answered but wrong signs, use of degrees and reversed coordinates were all seen in part (b)(ii). Part (b)(iii) was frequently not attempted or marks were lost due to an apparently careless omission of brackets. Sketches for $\sin 2x$ resulted in a spread of marks awarded. Those sketches gaining no credit usually resulted from stretching the given graph of $\sin x$ with a scale factor 2 parallel to the y -axis.

One mark was the usual award for those stretching the graph of $\sin x$ parallel to the x -axis but with scale factor 2. Those who correctly stretched $\sin x$ parallel to the x -axis with scale

factor $\frac{1}{2}$ scored at least two marks, and more often at least four marks, with a significant number failing to get the fifth mark because the x -coordinates of the maximum points were not shown.

Q30.

Part (a) was well answered by the majority of candidates. $\sec x$ being changed to $\sin x$ was a common error. The first angle was usually found correctly but the second angle was often incorrect, with 318 being a common error. There were very few cases of angles in radians being found.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was the use of $\tan^2 x = \sec^2 x + 1$, but $5 \sec x = 5(\tan x + 1)$ was also seen. For those who successfully reached the correct quadratic, the factorisation, or use of the quadratic formula, was usually well done, though there were some sign errors. For the final four answers, many totally correct solutions were seen. A common error at this stage was 284 degrees, from $180 + 104$, and not the correct answer of 256. There were a number of candidates who failed to correctly round their answers to the nearest degree, with $104.47 = 104.5 = 105$ seen.

Q31.

In part (a), nearly all candidates quoted the $\sin 2x$ expansion correctly, and attempted to use it in part (b). In part (b), many cancelled the $\cos x$ terms without considering $\cos x = 0$, and so lost a mark. Some candidates, in dividing by $\cos x$, brought $\tan x$ into the equation

and scored no further marks. Most candidates went on to find $\sin^{-1}\left(-\frac{1}{3}\right)$, although a few made a sign error. The calculator solution of -17.5° was interpreted correctly by most candidates, to give solutions in the required range. The responses to the 'proof in part (b)' varied considerably. Some candidates seemed to see immediately that they needed the $1 - 2 \sin^2 x$ version of $\cos 2x$ and obtained the requested result very concisely, and it was nice to see them justifying division by $\sin x$.

Others, although not making any mistakes, took rather longer to arrive at the result, some replacing 1 with $\sin^2 x + \cos^2 x$. Some candidates made no progress beyond quoting $\cos 2x = \cos^2 x - \sin^2 x$. For those who did make further progress, failure to complete was usually due to a dropped 2 or 1, or just losing sight of what they were trying to do. Those few candidates who worked with just the left hand side of the expression usually could not

establish the result.

Q32.

This question which tested trigonometrical identities and solutions of trigonometrical equations was by far the worse answered question on the paper. In part (a) candidates

generally could recall the identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ but proofs to reach $\tan \theta = 5$ were not always convincing. Those who started with $\sin \theta - \cos \theta = 4 \cos \theta$ were generally more successful than those who started by dividing the numerator by $\cos \theta$.

In part (b)(i) candidates could generally quote the relevant identity, $\cos^2 x + \sin^2 x = 1$, but could rarely use it in a convincing manner.

In the final part of the question, although the majority of candidates could see the use of the work in part (b)(i), a significant number of candidates could not solve the quadratic equation in $\sin x$. In some cases candidates just solved the equation by removing 'sin' wherever it occurred. Better attempts reached $2\sin x = 1$ and $\sin x = -1$ but then it was not uncommon to see ' $2\sin x = 1$ so $2x = 90^\circ$, $x = 45^\circ$, 225° ' and ' $\sin x = -1$ so no solution'.

Q33.

Most candidates obtained the method mark in part (a) by obtaining -0.32 . Many candidates went on to complete this part correctly, although incorrectly writing their answer to 2dp was a common error. Answers in degrees were not common, but they were seen.

Part (b) was answered very well by the majority of candidates with the correct trigonometric identity being used.

In part (c) most candidates were able to successfully factorise the quadratic expression, and many went on to complete the question correctly. Marks were lost by 'extra' values being given or poor accuracy of writing their answers to 2dp.

Q34.

There were many correct answers to part (a)(i). Rather than using $\cos 2x = 2\cos^2 x - 1$ directly, some answers were obtained from $\cos 2x = \cos^2 x - \sin^2 x = \cos^2 x - (1 - \cos^2 x)$. This sometimes led to sign errors. Other errors were the use of $\cos^2 x - 1$ or failure to multiply the -1 by 3.

In part (a)(ii) most candidates factorised successfully but then wasted time solving for x . The question did not ask for solutions. Those who used the quadratic formula often made mistakes.

Candidates generally demonstrated that they had both the knowledge and appropriate skill to answer part (b)(i) successfully and there was a pleasing number of correct solutions. The most common error was to write $\tan \alpha = 7/3$ instead of $3/7$.

In part (b)(ii) not all candidates read the request to give answers correct to the nearest 0.1° . These candidates were only penalised once. A common incorrect answer was 171.5 from $180 - 8.5$ instead of $180 - 31.7 - 23.2$. Clearly there was confusion when dealing

with the obtuse angle. Some candidates changed $\sin(\theta + \alpha)$ to $\sin(\theta - \alpha)$, so could not complete successfully.

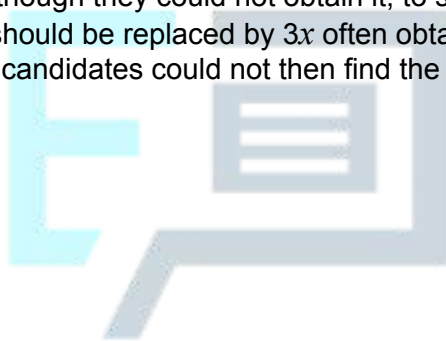
In part (c)(i) most candidates did not realise the need to use exact values and they used a calculator to find angle β and then its cosine. This gained no credit. Those who drew a diagram fared better as they often went on to use Pythagoras' theorem in an appropriate form, which showed where the value of $1/3$ came from.

Marks could only be earned in part (c)(ii) if $\sin 2\beta = 2\sin\beta\cos\beta$ was seen. Some candidates used calculators once more and did not give an exact answer. Some did realise that 0.44444 was $4/9$ but didn't gain any credit unless the double angle sine formula had been used. Many candidates made the mistake of apparently thinking $\sin 2\beta$ was $2\sin\beta$.

Q35.

In general candidates found this question difficult, and average-grade candidates rarely scored more than a couple of marks. In part (a), the mark for knowing the correct trigonometrical identity was frequently gained by those who attempted the question, but then many floundered after making the common error " $3 + 1 - \cos^2\theta = 3 - 3\cos^2\theta$ ".

It was encouraging to find a higher proportion of candidates than usual using the printed answer from part (a), even though they could not obtain it, to start their solution to part (b). Those who realised that θ should be replaced by $3x$ often obtained the 40° solution, but a significant number of these candidates could not then find the remaining two solutions in the given interval.



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